

CY 248: Molecular Physical Chemistry for Engineers

Course Code	CY 248
Course Name	Molecular Physical Chemistry for Engineers
L-T-P-C	3-0-0-3
Prerequisites	None.
Intended for	B. Tech. 2nd year
Distribution	Elective
Approval	6th Senate (May 2014)

Course Contents

- **Brief Review of Elementary Thermodynamics**
Introduction, Application of Gibbs free energy, Calculation of an Equilibrium Constant that cannot be Measured Conveniently. [4 Lectures]
- **Introduction to Quantum Theory & Applications**
Introduction, one dimensional problems, Observing Vibrations and Rotations of Molecules by Spectroscopy, Infrared Spectroscopy, Electronic Excitations in Molecules. [10 Lectures]
- **Statistical Mechanics Fundamental Ideas and Applications**
Introduction, Statistical Occupation of Energy Levels, Boltzmann Distribution Function, Ensembles, Molecular Partition Function, Connecting the Molecular Partition Function to the Internal Energy and Entropy, Boltzmann Law on entropy, Applications of the Partition Function to Chemical Thermodynamics Problems. [12 Lectures]
- **Chemical Reactions**
Introduction, Collision Theory, Reactive Hard-Sphere Molecules, Transition State Theory of Chemical Reaction Rates, Connection of Transition State Theory to Collision Theory, unimolecular reactions, RRKM theory. [16 Lectures]

Text Books

1. John T. Yates and J. Karl Johnson, **Molecular Physical Chemistry for Engineers.**

References

1. K. A. McLauchlan, **Molecular Physical Chemistry - A Concise Introduction**
2. D. Mcquarrie and J. Simon, **Physical Chemistry: A Molecular Approach.**
3. P. M. S. Monk, **Physical Chemistry: Understanding our Chemical World.**
4. S. Warren, **The Physical Basis of Chemistry.**
5. R. J. Silbey, R. A. Alberty, M. G. Bawendi, **Physical Chemistry**
6. P. Atkins and J. de Paula, **Physical Chemistry.**